PRACTICAL - 05

Part A — Practice Programs (Hour 1)

PROBLEM1	**Shape areas: an abstract Shape with abstract area(); subclasses Circle, Rectangle, Triangle. Put a mix in a Shape[] and loop once to print each area and the running total, tracking the largest — one loop handling every shape through polymorphism.**
CODE	

```
abstract class Shape {

   public abstract double area();
}

class Circle extends Shape {

   private double radius;

   public Circle(double radius) {
      this.radius = radius;
   }

   public double area() {
      return Math.PI * radius * radius;
   }
}

class Rectangle extends Shape {

   private double width, height;

   public Rectangle(double width, double height) {
      this.width = width;
      this.height = height;
   }

   public double area() {
      return width * height;
   }
}

class Triangle extends Shape {

   private double base, height;
```

```
    public Triangle(double base, double height) {
        this.base = base;
        this.height = height;
    }

    public double area() {
        return 0.5 * base * height;
    }
}

public class ShapeTest {

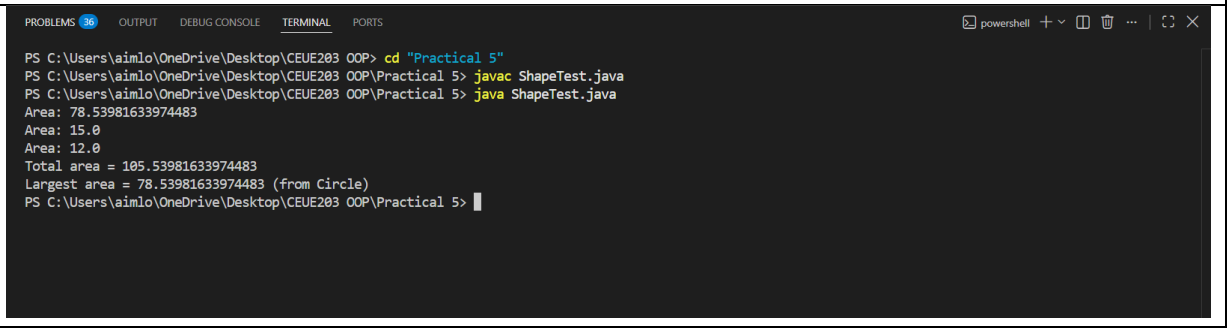
    public static void main(String[] args) {
        Shape[] shapes = {
            new Circle(5),
            new Rectangle(3, 5),
            new Triangle(4, 6),};
        double totalArea = 0;
        double largestArea = Double.MIN_VALUE;
        Shape largestShape = null;

        for (Shape s : shapes) {
            double area = s.area();
            System.out.println("Area: " + area);
            totalArea += area;

            if (area > largestArea) {
                largestArea = area;
                largestShape = s;
            }
        }

        System.out.println("Total area = " + totalArea);
        System.out.println("Largest area = " + largestArea + " (from " +
largestShape.getClass().getSimpleName() + ")");
    }
}
```

OUTPUT



PROBLEM2	**Payroll: an abstract Employee with abstract monthlySalary(); FullTime (fixed), PartTime (hours × rate) and Intern (stipend), each calling super for the shared name/id. Loop a mixed array to print each salary and the total; use instanceof to add a note only for interns.**
CODE	

```
abstract class Employee {

   private String name;
   private int id;

   public Employee(String name, int id) {
      this.name = name;
      this.id = id;
   }

   public String getName() {
      return name;
   }

   public int getId() {
      return id;
   }

   public abstract double monthlySalary();
}

class FullTime extends Employee {

   private double salary;

   public FullTime(String name, int id, double salary) {
      super(name, id);
      this.salary = salary;
   }

   public double monthlySalary() {
      return salary;
   }
}

class PartTime extends Employee {

   private int hours;
   private double rate;
```

```
    public PartTime(String name, int id, int hours, double rate) {
        super(name, id);
        this.hours = hours;
        this.rate = rate;
    }

    public double monthlySalary() {
        return hours * rate;
    }
}

class Intern extends Employee {

    private double stipend;

    public Intern(String name, int id, double stipend) {
        super(name, id);
        this.stipend = stipend;
    }

    public double monthlySalary() {
        return stipend;
    }
}

public class PayrollTest {

    public static void main(String[] args) {
        Employee[] employees = {
            new FullTime("Om", 72, 500000),
            new PartTime("Jugal", 65, 80, 200),
            new Intern("Yug R", 62, 5000),
            new FullTime("Yug T", 75, 60000),};

        double total = 0;

        for (Employee e : employees) {
            double salary = e.monthlySalary();
            System.out.print(e.getName() + " (ID " + e.getId() + ") salary is " + salary);
            if (e instanceof Intern) {
                System.out.print(" [Internship]");
            }
            System.out.println();
            total += salary;
        }

        System.out.println("Total payroll = " + total);
```

```
    }
}
```

OUTPUT



PROBLEM3 **(For Advanced Learners – home) Model a few media types that each compute a late fee differently and total the fees for a returned batch. Design the hierarchy and rules yourself.**

CODE

```
abstract class Media {
    private String title;
    private int id;

    public Media(String title, int id) {
        this.title = title;
        this.id = id;
    }

    public String getTitle() { return title; }
    public int getId() { return id; }

    public abstract double lateFee(int daysLate);
}

class Book extends Media {
    public Book(String title, int id) {
        super(title, id);
    }

    @Override
    public double lateFee(int daysLate) {
        return daysLate * 2.0;
    }
}

class DVD extends Media {
    public DVD(String title, int id) {
        super(title, id);
    }
```

```
    public double lateFee(int daysLate) {
        return daysLate * 5.0;
    }
}

class Magazine extends Media {
    public Magazine(String title, int id) {
        super(title, id);
    }


    public double lateFee(int daysLate) {

        return daysLate > 0 ? 10.0 : 0.0;
    }
}

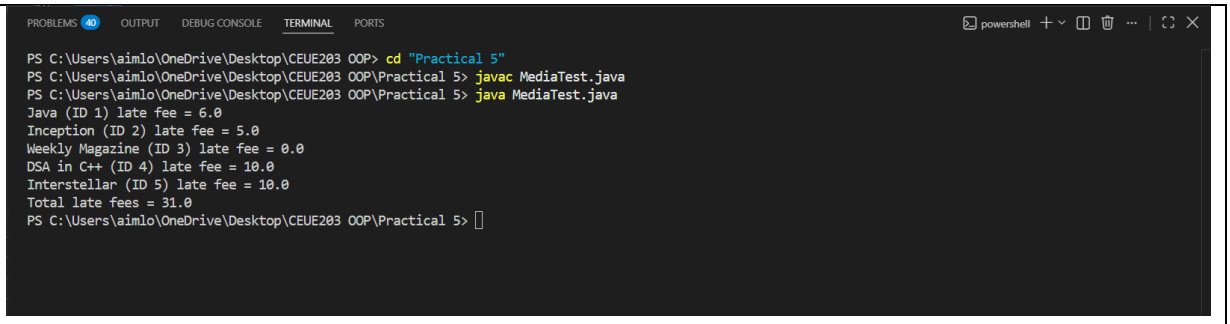
public class MediaTest {
    public static void main(String[] args) {
        Media[] items = {
            new Book("Java", 1),
            new DVD("Inception", 2),
            new Magazine("Weekly Magazine", 3),
            new Book("DSA in C++", 4),
            new DVD("Interstellar", 5)
        };

        int[] daysLate = {3, 1, 0, 5, 2};
        double totalFee = 0;

        for (int i = 0; i < items.length; i++) {
            double fee = items[i].lateFee(daysLate[i]);
            System.out.println(items[i].getTitle() + " (ID " + items[i].getId() +
                    ") late fee = " + fee);
            totalFee += fee;
        }

        System.out.println("Total late fees = " + totalFee);
    }
}
```

OUTPUT


PROBLEMS 40
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP> cd "Practical 5"
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> javac MediaTest.java
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5> java MediaTest.java
Java (ID 1) late fee = 6.0
Inception (ID 2) late fee = 5.0
Weekly Magazine (ID 3) late fee = 0.0
DSA in C++ (ID 4) late fee = 10.0
Interstellar (ID 5) late fee = 10.0
Total late fees = 31.0
PS C:\Users\aimlo\OneDrive\Desktop\CEUE203 OOP\Practical 5>